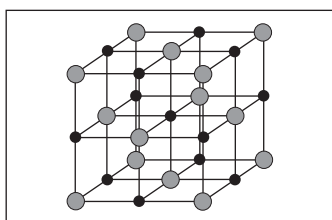
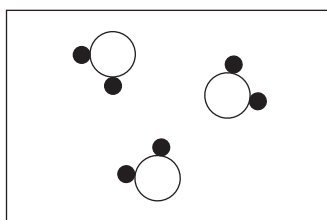


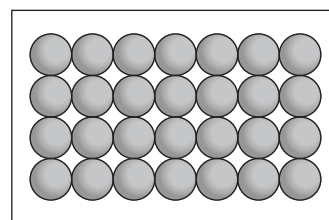
3. The following diagrams show the structure of substances.



A (solid)



B (gas)



C (solid)

- (a) State which of the structures (**A**, **B** or **C**) is an ionic compound. Give **one** reason for your answer. [2]

Letter:

Reason:

.....
.....

- (b) Sodium chloride is an ionic compound which consists of sodium and chloride ions.

Explain why sodium chloride will not conduct electricity when it is a solid but will conduct electricity when it is dissolved in water. [2]

.....
.....



- (c) Some of the materials used in sporting equipment are shown in the table below.

Material	Thermal conductivity (units)	Melting point (°C)	Density (g/cm ³)	Tensile strength (MPa)
carbon fibre	21	3650	1.78	4137
aluminium	210	660	2.71	276
lead	35	327	11.3	12

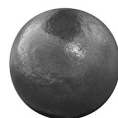
Use the data in the table to answer the following questions.

- (i) Carbon fibre is an example of a composite material used to make bicycle frames for use in cycle races.

State **two** reasons why modern bicycle frames are made of carbon fibre rather than aluminium. [2]

1.
2.

- (ii) A shot put contains lead.



Use the equation

$$\text{mass} = \text{density} \times \text{volume}$$

to find the mass of lead in a shot put that has a volume of 700 cm³. [2]

mass = g

